ГЕОМЕТРИЯ. Инварианты. Матрица.

Общее ур-е: $a_{11}x^2+2a_{12}xy+a_{22}y^2+2a_{13}x+2a_{23}y+a_{33}=0$

$$I(\lambda)=\begin{vmatrix} a_{11}-\lambda & a_{12} \\ & a_{22}-\lambda \end{vmatrix}$$ ← инвариант

$$\tilde a_{11}\tilde x^2+\tilde a_{22}\tilde y^2+\tilde a_{33}=0,$$

$$I_2=\tilde a_{11}\tilde a_{22}$$

$$I_3=\tilde a_{11}\tilde a_{22}\tilde a_{33}=I_2\tilde a_{33}\Rightarrow \tilde a_{33}=\frac{I_3}{I_2}.$$

$$I(\lambda)=(\tilde a_{11}-\lambda)(\tilde a_{22}-\lambda),$$
$$I(\tilde a_{11})=I(\tilde a_{22})=0.$$

$\tilde a_{11}\tilde x^2+2\tilde a_{23}\tilde y=0$: $\quad I_1=\tilde a_{11}$

ур-ие парабол $\quad I_2=0$

$$0\neq I_3=\begin{vmatrix}\tilde a_{11} & 0 & 0\\ 0 & 0 & a_{23}\\ 0 & a_{23} & 0\end{vmatrix}=\tilde a_{11}\cdot\left(-\tilde a_{23}^2\right)=-I_1\tilde a_{23}^2;$$

$$\tilde a_{23}=\pm\sqrt{-\frac{I_3}{I_1}}.$$

I_2?

$I_2=0$ $\quad$ $I_2\neq0$

I_3? $\quad$ I_3? → $J(\lambda)=0$

$I_3=0$ $\quad$ $I_3\neq0$ $\quad$ Гип-ла, Эл., ПХ.

$(\tilde a_{11}\tilde x^2 + \tilde a_{33} = 0)$

$\perp\ J_2 = 0$

$$\begin{pmatrix} a_{11} & 0 & 0 \\ & 0 & 0 \\ & & a_{33} \end{pmatrix} = J_3 = 0$$

$\downarrow$

ПП/мнимые ПП

$J(\lambda) = 0$

пар-ла

$$\begin{vmatrix} a_{11} & a_{13} \\ & a_{33} \end{vmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ & a_{33} \end{vmatrix} = J_2^* . \leftarrow$$ величина инварианта при повороте СК.

откуда-то из прошлого

$$\begin{cases} \tilde a_{11} = a_{11}\cos^2\varphi + 2a_{12}\cos\varphi\sin\varphi + a_{22}\sin^2\varphi, \\ \tilde a_{22} = a_{11}\sin^2\varphi - 2a_{12}\cos\varphi\sin\varphi + a_{22}\cos^2\varphi, \\ \tilde a_{13} = a_{13}\cos\varphi + a_{23}\sin\varphi, \\ \tilde a_{23} = a_{13}\sin\varphi + a_{23}\cos\varphi \end{cases}$$

$$(\tilde a_{11} + \tilde a_{22})\tilde a_{33} - (\tilde a_{13}^2 + \tilde a_{23}^2) = (a_{11} + a_{22})a_{33} - (a_{13}^2 c^2 \cancel{+ 2a_{13}a_{23}sc} + a_{23}^2 s^2 +$$

$$+ a_{13}^2 s^2 \cancel{- 2a_{13}a_{23}sc} + a_{23}^2 c^2) =$$

$$= (a_{11} + a_{22})a_{33} + (a_{13}^2 + a_{23}^2).$$

$$J_2^* = \tilde a_{11}\tilde a_{33} = J_1 \tilde a_{33} \Rightarrow \boxed{\tilde a_{33} = \frac{J_2^*}{J_1}.}$$

При переносе СК:

$\tilde a_{11} = a_{11}$

$\tilde a_{22} = a_{22}$

$\tilde a_{13} = a_{13} + a_{11}x_0 + a_{12}y_0$

$\tilde a_{23} = a_{23} + a_{12}x_0 + a_{22}y_0$

$\tilde a_{33} = (a_{11} + a_{22})(a_{33} + \dots) - (a_{13}^2 + a_{23}^2 + \dots) = F(x_0, y_0).$

$a_{11} a_{22} - a_{12}^2 = 0 \leftarrow J_2 = 0, \ J_3 = 0.$

$x_0^2 : \cancel{a_{11}^2} - \cancel{a_{11}^2} + \underline{a_{22} a_{11} - a_{12}^2} = 0$

$x_0 y_0 : 2\cancel{a_{11}} a_{12} - 2\cancel{a_{11}} a_{12} + 2\cancel{a_{22}} a_{12} - 2\cancel{a_{12}} a_{22} = 0$

$y_0^2 : \underline{a_{11} a_{22} - a_{12}^2} \cancel{+ a_{22}^2} \cancel{- a_{22}^2} = 0$

$x_0 : 2a_{11} a_{13} - 2a_{11} a_{13} + 2a_{22} a_{13} - 2a_{23} a_{12} = 2(a_{22} a_{13} - a_{23} a_{12})$

$J_3 = \begin{vmatrix} a_{11} & \cdots \\ \vdots & \end{vmatrix} =$